

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 6 — Trigonometry and Bearing

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

1. Write your name, roll number, and section clearly on the answer sheet.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B** (Short Questions): Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	There are how many minutes in one degree?	30	60	90	3600	○○○○
2	The measure of a complete angle in circular system is:	π rad	2π rad	3π rad	4π rad	○○○○
3	The length of a circular arc is given by:	$l = r + \theta$	$l = r / \theta$	$l = r\theta$	$l = \theta / r$	○○○○
4	The value of $\sin 30^\circ$ is:	$\sqrt{3} / 2$	$1 / 2$	$1 / \sqrt{2}$	1	○○○○
5	In the second quadrant, which ratio is positive?	$\cos \theta$	$\tan \theta$	$\sin \theta$	$\sec \theta$	○○○○
6	Which Pythagorean identity is correct?	$\sin^2\theta - \cos^2\theta = 1$	$\sec^2\theta + \tan^2\theta = 1$	$\sin^2\theta + \cos^2\theta = 1$	$\operatorname{cosec}^2\theta + \cot^2\theta = 1$	○○○○
7	$\cos\theta = b/c$ in a right triangle represents:	Perp / Hyp	Base / Hyp	Hyp / Base	Perp / Base	○○○○
8	$50^\circ 30'$ expressed in decimal degrees is:	50.2°	50.3°	50.4°	50.5°	○○○○
9	The angle between the horizontal and line of sight to an object <i>above</i> the horizontal is:	Depression	Elevation	Vertical angle	Quadrantal angle	○○○○
10	Bearings are measured clockwise from:	South	East	North	West	○○○○
11	The bearing of due east direction is:	000°	045°	090°	180°	○○○○
12	If $\cos x = \sin x$, the value of x is:	30°	45°	60°	90°	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt ALL 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Convert $45^\circ 45' 45''$ into seconds.	3	OR	Convert $216000''$ into $D^\circ M'S''$.	3
2(ii)	Convert 45° into radians (express in terms of π and as a decimal, use $\pi \approx 3.1416$).	3	OR	Convert $3\pi/4$ radians into $D^\circ M'S''$.	3
2(iii)	Find arc length and area of sector: $r = 12$ m, $\theta = 120^\circ$.	3	OR	A 30-inch pendulum swings through 30° . Find arc length through which the tip swings.	3
2(iv)	Find all six trigonometric ratios of θ if $\sin\theta = \sqrt{3}/2$ and terminal ray is in the first quadrant.	3	OR	Write all six trigonometric ratios of 45° .	3
2(v)	Evaluate: $\sin 60^\circ - \cos 30^\circ$.	3	OR	Evaluate: $\sec 45^\circ \operatorname{cosec} 45^\circ - \sin 30^\circ \cos 60^\circ$.	3
2(vi)	Terminal ray of θ is in the first quadrant and $\cos\theta = 1/2$. Find the remaining five trigonometric ratios.	3	OR	Terminal ray of θ is in the second quadrant and $\operatorname{cosec}\theta = 2$. Find the value of $\cos\theta \cdot (\sin^2\theta - \cos^2\theta) / (\sin^2\theta + \cos^2\theta)$.	3
2(vii)	Prove: $(1 - \sin^2\theta) \sec^2\theta = 1$.	3	OR	Prove: $1/(\sec\theta - \tan\theta) = \sec\theta + \tan\theta$.	3

2(viii)	Solve $\triangle ABC$: $\angle C=90^\circ$, $\angle A=30^\circ$, $a=6$ cm. Find $\angle B$, b and c .	3	OR	Solve $\triangle PQR$: $\angle Q=90^\circ$, $\angle P=60^\circ$, $PQ=4\sqrt{3}$ cm. Find $\angle R$, QR and PR .	3
2(ix)	From a point 20 m from a tree, angle of elevation of the treetop is 30° . Find the height of the tree.	3	OR	Shadow of a 10 m pole is $10\sqrt{3}$ m long. Find the angle of elevation of the sun.	3
2(x)	Define bearing. Write the three-figure bearings of north, south, east and west.	3	OR	Bearing of boat B from boat A is 250° . Find the bearing of boat A from boat B.	3
2(xi)	A car leaves a garage at bearing 040° and travels 13 km. How many km north and how many km east has it travelled?	3	OR	A plane is 119 km west and 100 km south of an airport. At what bearing should the pilot fly to return to the airport?	3

SECTION C — Long Questions ($4 \times 5 = 20$ Marks)

Attempt ALL 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Find arc length and area of sector for $r = 6$ dm, $\theta = 60^\circ 45' 30''$. Then verify the area using $A = \frac{1}{2}r^2\theta$.	5	OR	Length of an arc is 4 cm and area of its sector is 16 cm^2 . Find the radius and the radian measure of the central angle using $l = r\theta$ and $A = \frac{1}{2}r^2\theta$.	5
3(ii)	Prove the identities: (a) $(\sin\theta - 2\sin 3\theta)/(2\cos 3\theta - \cos\theta) = \tan\theta$ (b) $\sqrt{(\sec^2\theta + \operatorname{cosec}^2\theta)} = \tan\theta + \cot\theta$	5	OR	Prove the identities: (a) $(1 - \sin\theta)/(1 + \sin\theta) = (\sec\theta - \tan\theta)^2$ (b) $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$	5
3(iii)	An aeroplane at altitude 900 m sees two ships sailing towards it, angles of depression 30° and 60° . Find the distance between the ships.	5	OR	From the top of a 120 m tower, angles of depression of two boats on the same side are 60° and 45° . Find the distance between the boats with full working.	5
3(iv)	A ship goes from A to B at bearing 040° for 35 km, then B to C at bearing 130° for 50 km. Find: (a) distance AC, (b) bearing of C from A.	5	OR	Babar Azam walks P to Q at bearing 050° for 3 km, then Q to R at bearing 140° . Distance $PR = 5$ km. Find: (a) distance QR, (b) bearing of R from P.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Ch 6 — Trigonometry and Bearing

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